

```
!pip install pandas
import pandas as pd
import numpy as np
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
```

```
print(pd.__version__)
```

```
2.2.2
```

```
#SERIES:-
s=pd.Series([1,2,3,4,5])
print(s)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
```

```
#FROM LIST
s=pd.Series([1,2,3,4,5],index=['a','b','c','d','e'])
print(s)
```

```
a    1
b    2
c    3
d    4
e    5
dtype: int64
```

```
#
student={
    "name":"dhanushka",
    "branch":"CSE"
}
se=pd.Series(student)
print(se)
```

```
name      dhanushka
branch      CSE
dtype: object
```

```
#operations
n=pd.series([1,2,3,4,5])
```

```
#arithmetic
s1=pd.Series([1,2,3,4,5])
s2=pd.Series([10,20,30,40,50])
print(s1+s2)
```

```
0    11
1    22
2    33
3    44
4    55
dtype: int64
```

```
#statistics
n=pd.series([1,2,3,4,5])
```

```
s.mean()
```

```
np.float64(3.0)
```

```
s.max()
```

```
5
```

```
s.sum()
```

```
np.int64(15)
```

```
#boolean  
n=pd.series([1,2,3,4,5])
```

```
#  
names=pd.Series(['dhanu','aswi','manu','akki'])  
names.str.lower()
```

```
      0  
0  dhanu  
1   aswi  
2   manu  
3   akki
```

dtype: object

```
names.str.upper()
```

```
      0  
0  DHANU  
1   ASWI  
2   MANU  
3   AKKI
```

dtype: object

```
names.str.len()
```

```
      0  
0    5  
1    4  
2    4  
3    4
```

dtype: int64

```
names.str.startswith('h')
```

```
      0  
0  False  
1  False  
2  False  
3  False
```

dtype: bool

```
str={  
    "names":["anu","laddu","dhanu"],  
    "branch":["cse","ece","eee"]  
}  
df=pd.DataFrame(str)  
print(df)
```

```
      names branch  
0      anu    cse  
1    laddu    ece  
2    dhanu    eee
```

```
arr=np.random.rand(3,4)  
c=pd.DataFrame(arr,columns=['a','b','c','d'])  
print(c)
```

	a	b	c	d
0	0.407706	0.376578	0.387869	0.486458
1	0.075967	0.684359	0.600945	0.050504
2	0.417398	0.865036	0.545865	0.066424

```
data={
    'name':['anu', 'pooji', 'lucky'],
    'age':[20,21,22],
    'score':[90,80,10],
    'grade':['A','B','C']
}
df=pd.DataFrame(data)
print(df)
```

	name	age	score	grade
0	anu	20	90	A
1	pooji	21	80	B
2	lucky	22	10	C

```
df.head()
```

	name	age	score	grade
0	anu	20	90	A
1	pooji	21	80	B
2	lucky	22	10	C

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 4 columns):
#   Column  Non-Null Count  Dtype
---  ---
0    name    3 non-null        object
1    age      3 non-null        int64
2    score    3 non-null        int64
3    grade    3 non-null        object
dtypes: int64(2), object(2)
memory usage: 228.0+ bytes
```

```
df.describe()
```

	age	score
count	3.0	3.000000
mean	21.0	60.000000
std	1.0	43.588989
min	20.0	10.000000
25%	20.5	45.000000
50%	21.0	80.000000
75%	21.5	85.000000
max	22.0	90.000000

```
df.columns
```

```
Index(['name', 'age', 'score', 'grade'], dtype='object')
```

```
df.index
```

```
RangeIndex(start=0, stop=3, step=1)
```

```
df.nunique
```

```
pandas.core.frame.DataFrame.nunique  
def nunique(axis: Axis=0, dropna: bool=True) -> Series
```

Count number of distinct elements in specified axis.

Return Series with number of distinct elements. Can ignore NaN values.

Parameters

```
df['name']
```

	name
0	anu
1	pooji
2	lucky

dtype: object

```
df[['name','score']]
```

	name	score
0	anu	90
1	pooji	80
2	lucky	10

```
df.loc[0:2]
```

	name	age	score	grade
0	anu	20	90	A
1	pooji	21	80	B
2	lucky	22	10	C

```
df.loc[0:2,['name','score']]
```

	name	score
0	anu	90
1	pooji	80
2	lucky	10

```
df.iloc[0]
```

	0
name	anu
age	20
score	90
grade	A

dtype: object

```
df.iloc[0:2]
```

	name	age	score	grade
0	anu	20	90	A
1	pooji	21	80	B

```
df.iloc[0:2,0:2]
```

	name	age
0	anu	20
1	pooji	21

```
df.iloc[-1]
```

	2
name	lucky
age	22
score	10
grade	C

```
dtype: object
```

```
df.loc[0, 'name']
```

```
'anu'
```

```
df.iloc[0,0]
```

```
'anu'
```

```
df[df['score']>80]
```

	name	age	score	grade
0	anu	20	90	A

```
df[(df['score']>80) & (df['grade']=='A')]
```

	name	age	score	grade
0	anu	20	90	A

```
df[(df['age']<24) & (df['score']>90)]
```

	name	age	score	grade
--	------	-----	-------	-------

```
#query()
```

```
df.query('score > 80 and age < 25')
```

	name	age	score	grade
0	anu	20	90	A

```
df.query('grade=="A"')
```

	name	age	score	grade
0	anu	20	90	A

```
df[df['name'].isin(['anu', 'lucky'])]
```

	name	age	score	grade
0	anu	20	90	A
2	lucky	22	10	C

```
df['grade']=df['score']>=80  
print(df)
```

	name	age	score	grade
0	anu	20	90	True
1	pooji	21	80	True
2	lucky	22	10	False

```
#modify existing column
df['score']=df['score']+5
print(df)
```

	name	age	score	grade
0	anu	20	95	True
1	pooji	21	85	True
2	lucky	22	15	False

```
def grade(score):
    if score >=90:
        return 'A'
    elif score >=60:
        return 'B'
    else:
        return 'C'
df['grade_letter']=df['score'].apply(grade)
print(df)
```

	name	age	score	grade	grade_letter
0	anu	20	95	True	A
1	pooji	21	85	True	B
2	lucky	22	15	False	C

```
result={
    True:'pass',
    False:'fail'
}
pas={'A':90,'B':75,'D':60}
df['pas']=df['grade'].map(pas)
df['result']=df['grade'].map(result)
print(df)
```

	name	age	score	grade	grade_letter	pas	result
0	anu	20	95	True	A	NaN	pass
1	pooji	21	85	True	B	NaN	pass
2	lucky	22	15	False	C	NaN	fail

```
df.drop('grade_letter',axis=1,inplace=True)
print(df)
```

	name	age	score	pas	result
0	anu	20	95	NaN	pass
1	pooji	21	85	NaN	pass
2	lucky	22	15	NaN	fail

```
#renamae
df.rename(columns={'score':'marks'},inplace=True)
print(df)
```

	name	age	marks	pas	result
0	anu	20	95	NaN	pass
1	pooji	21	85	NaN	pass
2	lucky	22	15	NaN	fail

```
df.rename(columns={'name':'students'},inplace=True)
print(df)
```

	students	age	marks	pas	result
0	anu	20	95	NaN	pass
1	pooji	21	85	NaN	pass
2	lucky	22	15	NaN	fail

```
df.replace('a',np.nan,inplace=True)
print(df)
```

	students	age	marks	pas	result
0	anu	20	95	NaN	pass
1	pooji	21	85	NaN	pass
2	lucky	22	15	NaN	fail

```
df.sort_values(['pas','marks'],ascending=[True,False])
```

	students	age	marks	pas	result
0	anu	20	95	NaN	pass
1	pooji	21	85	NaN	pass
2	lucky	22	15	NaN	fail

```
df.sort_index(ascending=False)
print(df)
```

	students	age	marks	pas	result
0	anu	20	95	NaN	pass
1	pooji	21	85	NaN	pass
2	lucky	22	15	NaN	fail

```
df.nsmallest(2, 'age')
print(df)
```

	students	age	marks	pas	result
0	anu	20	95	NaN	pass
1	pooji	21	85	NaN	pass
2	lucky	22	15	NaN	fail

```
employees = pd.DataFrame({
    'emp_id': [1, 2, 3, 4],
    'name': ['Alice', 'Bob', 'Charlie', 'Diana'],
    'dept_id': [10, 20, 10, 30]
})
departments = pd.DataFrame({
    'dept_id': [10, 20],
    'dept_name': ['Engineering', 'HR']
})
```